CS 331
Spring 2026

Craig Gotsman

Chapter 1

Database System Concepts and Architecture

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EWR flight departures - chronological

Destination	Departure	Flight	Airline	Terminal	Status
Bermuda (BDA)	12:00 pm	2T241	BermudAir		Landed - Delayed [+]
Boston (BOS)	12:00 pm	UA366	United Airlines	C	Landed - On-time [+]
Houston (IAH)	12:00 pm	NK1257	Spirit Airlines	B	Landed - On-time [+]
Los Angeles (LAX)	12:00 pm	UA432	United Airlines	C	En Route - On-time [+]
		CA7359	Air China		
		NZ9533	Air New Zealand		
		VA8144	Virgin Australia		
San Francisco (SFO)	12:00 pm	UA1343	United Airlines	C	En Route - Delayed [+]
		NZ9133	Air New Zealand		
		VA8162	Virgin Australia		
Toronto (YYZ)	12:00 pm	UA3400	United Airlines	C	Landed - On-time [+]
		AC3097	Air Canada		
Columbia (CAE)	12:02 pm	UA4507	United Airlines	C	Landed - On-time [+]
Phoenix (PHX)	12:03 pm	UA564	United Airlines	A	En Route - On-time [+]
Las Vegas (LAS)	12:05 pm	UA2050	United Airlines	A	En Route - On-time [+]
Charlotte (CLT)	12:17 pm	NK2186	Spirit Airlines	B	Landed - On-time [+]
Atlanta (ATL)	12:30 pm	UA1359	United Airlines	A	Landed - On-time [+]
		AC3023	Air Canada		
Houston (IAH)	12:30 pm	UA317	United Airlines	C	En Route - On-time [+]
		EK6065	Emirates		
		NZ6600	Air New Zealand		
Orlando (MCO)	12:30 pm	UA361	United Airlines	A	

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EWR flight departures – American Airlines

Destination	Departure	Flight	Airline	Terminal	Status
Miami (MIA)	05:21 am	AA1196	American Airlines	A	Landed - On-time [+]
Charlotte (CLT)	06:00 am	AA2991	American Airlines	A	Landed - On-time [+]
Dallas (DFW)	06:00 am	AA2728	American Airlines	A	Landed - On-time [+]
Phoenix (PHX)	06:30 am	AA2938	American Airlines	A	Landed - On-time [+]
Chicago (ORD)	07:00 am	AA4548	American Airlines	A	Landed - On-time [+]
Miami (MIA)	07:30 am	AA1370	American Airlines	A	Landed - On-time [+]
Charlotte (CLT)	08:00 am	AA1575	American Airlines	A	Landed - On-time [+]
Dallas (DFW)	08:10 am	AA1603	American Airlines	A	Landed - On-time [+]
Charlotte (CLT)	09:59 am	AA1724	American Airlines	A	Land
Phoenix (PHX)	10:20 am	AA2133	American Airlines	A	Land
Chicago (ORD)	11:53 am	AA3848	American Airlines	A	Land
Dallas (DFW)	12:37 pm	AA1700	American Airlines	A	En Ro
Miami (MIA)	12:39 pm	AA1692	American Airlines	A	Land
Charlotte (CLT)	01:30 pm	AA3204	American Airlines	A	Land
Chicago (ORD)	01:52 pm	AA3355	American Airlines	A	En Ro
Phoenix (PHX)	04:59 pm	AA1441	American Airlines	A	Sched
Charlotte (CLT)	05:10 pm	AA1216	American Airlines	A	Sched
San Francisco (SFO)	05:30 pm	AA7453	American Airlines	B	Sched
Dublin (DUB)	05:45 pm	AA8334	American Airlines	B	Sched
Dallas (DFW)	05:59 pm	AA1697	American Airlines	A	Sched

EWR flight departures – Terminal B

Destination	Departure	Flight	Airline	Terminal	Status
Houston (IAH)	12:00 pm	NK1257	Spirit Airlines	B	Landed - On-time [+]
Charlotte (CLT)	12:17 pm	NK2186	Spirit Airlines	B	Landed - On-time [+]
Myrtle Beach (MYR)	12:45 pm	NK1412	Spirit Airlines	B	Landed - On-time [+]
Fort Lauderdale (FLL)	01:00 pm	NK1491	Spirit Airlines	B	Landed - On-time [+]
Charleston (CHS)	02:00 pm	NK3002	Spirit Airlines	B	Landed - On-time [+]
Toronto (YTZ)	02:45 pm	PD2132	Porter Airlines	B	Landed - On-time [+]
		TS7443	Air Transat		
Orlando (MCO)	03:30 pm	NK2197	Spirit Airlines	B	En Route - Delayed [+]
Ottawa (YOW)	03:45 pm	PD2338	Porter Airlines	B	En Route - On-time [+]
		TS7094	Air Transat		
Dallas (DFW)	04:15 pm	NK2223	Spirit Airlines	B	Scheduled - Delayed [+]
Raleigh/Durham (RDU)	04:19 pm	NK254	Spirit Airlines	B	Scheduled - On-time [+]
Seattle (SEA)	04:36 pm	AS280	Alaska Airlines	B	Scheduled - On-time [+]
		HA4899	Hawaiian Airlines		
		JL6381	JAL		
		KE6271	Korean Air		
		TN2568	Air Tahiti Nui		
Miami (MIA)	05:00 pm	NK2890	Spirit Airlines	B	Scheduled - On-time [+]
Paris (CDG)	05:05 pm	AF63	Air France	B	Scheduled - On-time [+]
		DL8758	Delta Air Lines		
		KL2035	KLM Royal Dutch Airlines		

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MD Bank

MD Cashback credit card statement

Jane Smith
1100–522 University Ave.
Toronto ON M5G 1W7

Questions?
Call 1 800 267-2332

View your statement online:
md.ca

1 **Statement date:** April 17, 2025
2 **Statement balance:** $4,099.01
3 **Statement period:** March 18, 2025–April 17, 2025
4 **Payment due date:** May 8, 2025

Statement details

5 Transaction date	6 Posting date	7 Account activity	Amount
19 March	21 March	Payment BNS	-$1,600.00
03 April	04 April	Payment BNS	-$2,000.00
		Total payment activity	**-$3,600.00**
16 March	18 March	Flight Centre Travel Toronto ON	$2,252.51
06 April	07 April	Ticketmaster Canada Toronto ON	$398.74
08 April	09 April	BNS ATM Mississauga ON	$24.00
10 April	12 April	Cash Advance Fee	$5.00
11 April	13 April	BNS Loan	$725.55
11 April	13 April	Loblaws Toronto, ON	$121.36
12 April	13 April	Amazon Toronto, ON	$301.26
13 April	15 April	Netflix Toronto, ON	$50.45
15 April	17 April	MLS Sports Toronto, ON	$1,278.36
15 April	17 April	Goodlife Fitness Toronto, ON	$850.00
16 April	16 April	Loblaws Toronto, ON	$274.28
17 April	17 April	Cash Interest Charge	$0.27

Interest rates

	Interest charged this statement	Annual interest rate	Daily interest rate
Purchases	$0.00	21.99%	0.06024 %
Cash advances	$0.27	22.97%	0.06293 %
Fees	$0.00	21.99%	0.06024 %

Account summary

8	Previous balance As of March 17, 2025	$1,417.23
9	+Purchases	$6,252.51
10	+Cash advances	$24.00
11	+Interest	$0.27
12	+Fees	$5.00
13	+Other charges	$0.00
14	-Payments - Thank you	$3,600.00
	-Other credits	$0.00
	Statement balance	$4,099.01
15	Past due amount	$0.00
	Overlimit amount	$0.00
16	Minimum payment	$90.17

Payment due date:
May 8, 2025

17	Limit	$14,600.00
18	Available	$10,500.99

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Order Date	Order ID	Title	Category	ASIN/ISBN	UNSPSC Code	Seller	Purchase Price
01/01/20	112-5248654	MIULEE Set of 2, 12x20 Pillow Inserts S	HOME_BED_AND_B	B07M8RTF3G	52120000	Miulee Home	$15.99
01/01/20	112-5248654	Krispy Kreme Classic Keurig Single-Ser	GROCERY	B0798LF9DX	50000000	Amazon.com	$12.94
01/01/20	112-5248654	MIULEE Pack of 2, Velvet Soft Soild De	HOME_BED_AND_B	B07JPR33GL	52121512	Miulee Home	$11.99
01/09/20	112-2797108	The ConfiDental - Pack of 5 Moldable M	BEAUTY	B075FV78W8	53131600	Little Pearn	$11.02
01/15/20	112-4798990	Krispy Kreme Classic Keurig Single-Ser	GROCERY	B0798LF9DX	50000000	Amazon.com	$12.94
01/18/20	113-4707562	Aurrako Non Slip Rug Pads for Hardwo	HOME_FURNITURE_	B07Y1PXCBM	56100000	Aurrako	$6.99
01/18/20	113-4707562	Odd Easy Coffee Pod Holder Glass Top	KITCHEN	B07X8FGRLB	52150000	Odd Easy	$23.99
01/18/20	113-4707562	Aurrako Original Area Rug Gripper Pad	HOME_FURNITURE_	B07R5J47HT	56100000	Aurrako	$15.99
01/21/20	112-0099159	SoftTouch 4681095N, Cherry Furniture	HARDWARE	B004670YOK	31162701	Amazon.com	$8.99
01/28/20	113-2485619	Krispy Kreme Classic Keurig Single-Ser	GROCERY	B0798LF9DX	50000000	Amazon.com	$14.99
01/28/20	113-8296863	Nylabone Double Action Dog Chew	PET_SUPPLIES	B000ER3QM8	10110000	Amazon.com	$7.25

amazon

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Types of Databases and Database Applications

Traditional: Numeric and Textual Databases

- Credit card transactions
- Hotel or airline reservations
- Real estate listings
- Library book catalogs
- Online purchases
- Store inventory
- Stock market trades
- Uber rides
- Deliveries

powered by: ORACLE

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Types of Databases and Database Applications

Modern Applications

- Multimedia databases
- Geographic information systems (GIS)
- Document repositories
- Biological and genomic databases

powered by: mongoDB

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Internet-based Databases

- Social networks store information about people, their posts and connections between them

powered by: neo4j

- Search engines organize web pages for searching purposes
- AI systems organize all types of data for mining purposes

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Basic Definitions

Data:

- Known and meaningful facts that can be recorded.

Database:

- A collection of related data.
- A database is designed, built, and populated with data for a specific purpose.
- It has an intended group of users and some preconceived applications in which these users are interested.

Mini-world:

- Some parts of the real world about which data is stored in a database.
  For example, student grades and transcripts at a university.

Database Management System (DBMS):

- A software system to facilitate the creation and maintenance of a computerized database.

Database System:

- The DBMS software together with the data itself. Sometimes, the applications are also included.

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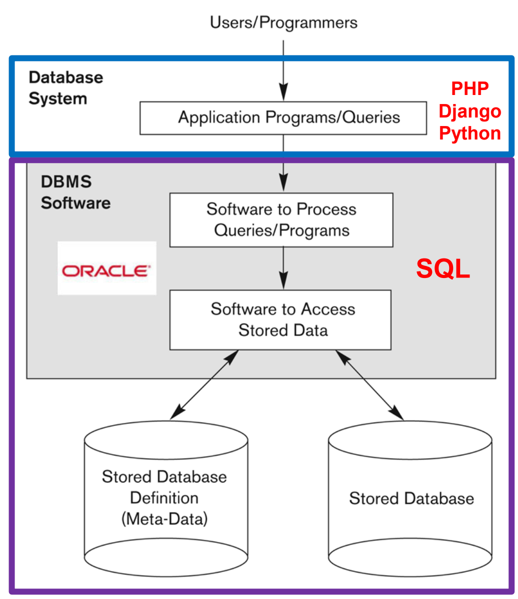
Database Actors

Users: Those who design, develop and maintain database applications, and those who actually use and control the database content.

Developers: Those who design and develop the DBMS software and related tools.



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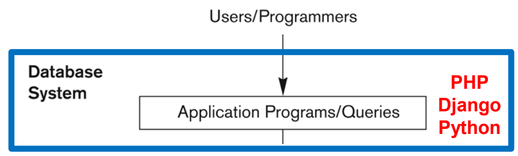
On the Scene



Database Administrators

Responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring efficiency of operations.

Database Designers

Responsible for defining the content, the structure, the constraints, and functions or transactions on the database. They must communicate with the end users and understand their needs.

End Users

Use the data for queries, reports and some of them update the database content.

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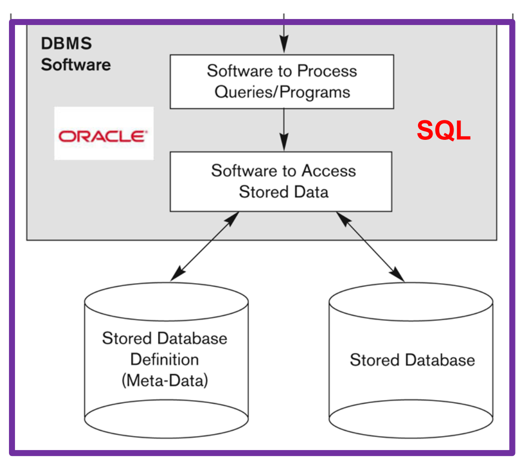
Behind the Scene

System Designers and Implementors: Design and implement DBMS packages in the form of modules and interfaces and test and debug them. The DBMS must interface with applications, language compilers, operating system components, etc.

Tool Developers: Design and implement software systems for modeling and designing databases, performance monitoring, prototyping, test data generation, user interface creation, simulation etc. that facilitate building of applications and allow using the database effectively.

Operators and Maintenance Personnel: Manage the actual running and maintenance of the database system hardware and software environment.



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Typical DBMS Functionality

- **Define** a particular database in terms of its data types, structures, and constraints
- **Construct** or Load the initial database contents on a secondary storage medium
- **Manipulate** the database:
  - Retrieval: Querying, generating reports
  - Modification: Insertions, deletions and updates to its content
  - Access the database through Web applications
- **Process** and **Share** by a set of concurrent users and application programs – keeping all data valid and consistent

- **Mini-world:**
  - Part of a UNIVERSITY environment.
- **Some mini-world entities:**
  - STUDENTs
  - COURSEs
  - SECTIONs (of COURSEs)
  - (academic) DEPARTMENTs
  - INSTRUCTORs
  - PREREQUISITEs
  - GRADEs

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Applications Against a Database

- Applications interact with a database by performing:
  - **Queries:** that access different parts of data and formulate the result of a request (read)
  - **Transactions:** that may read some data and "update" certain values or generate new data and store that in the database (write)
- Applications must not allow unauthorized users to access data
- Applications must keep up with changing user requirements against the database

- **Queries:**
  - Which STUDENTS study COURSE CS311 ?
  - Which INSTRUCTORS teach more than one COURSE ?
- **Transactions:**
  - Transfer the STUDENT John Doe from COURSE CS311 to COURSE CS345.
  - Add another SECTION to COURSE CS100.

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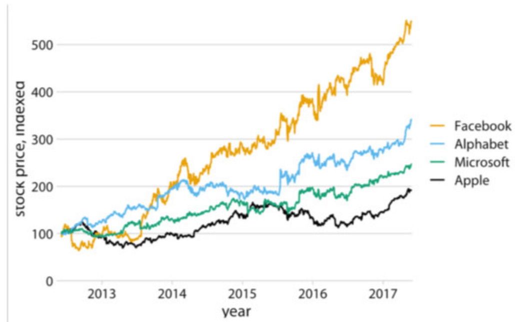
Additional DBMS Functionality

The DBMS may additionally provide:

- Protection or security measures to prevent unauthorized access
- “Active” processing to take internal actions on data
- Presentation and visualization of data
- Maintenance of the database and associated programs over the lifetime of the database application



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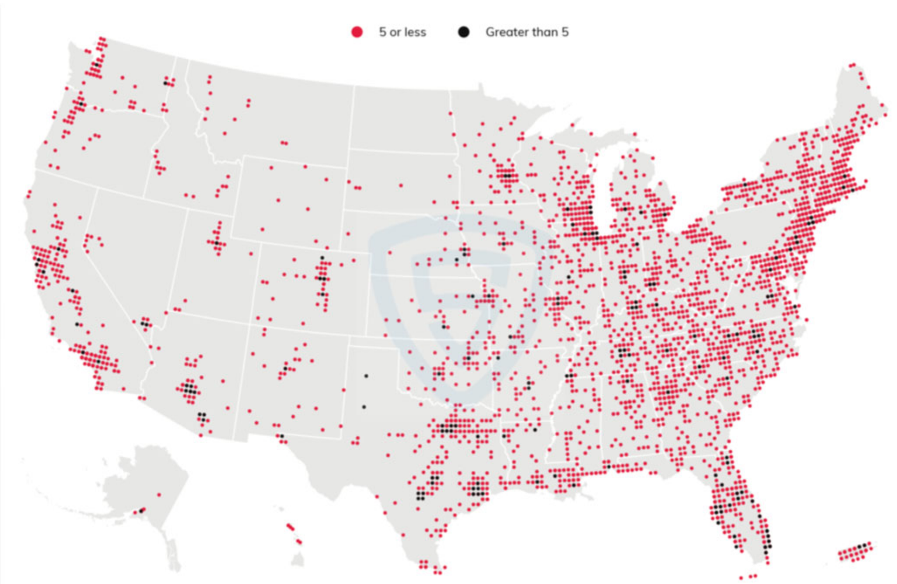

Walgreens pharmacy locations in the USA
Each grid point covers 10-mile radius with at least one location
Source: ScrapeHero.com

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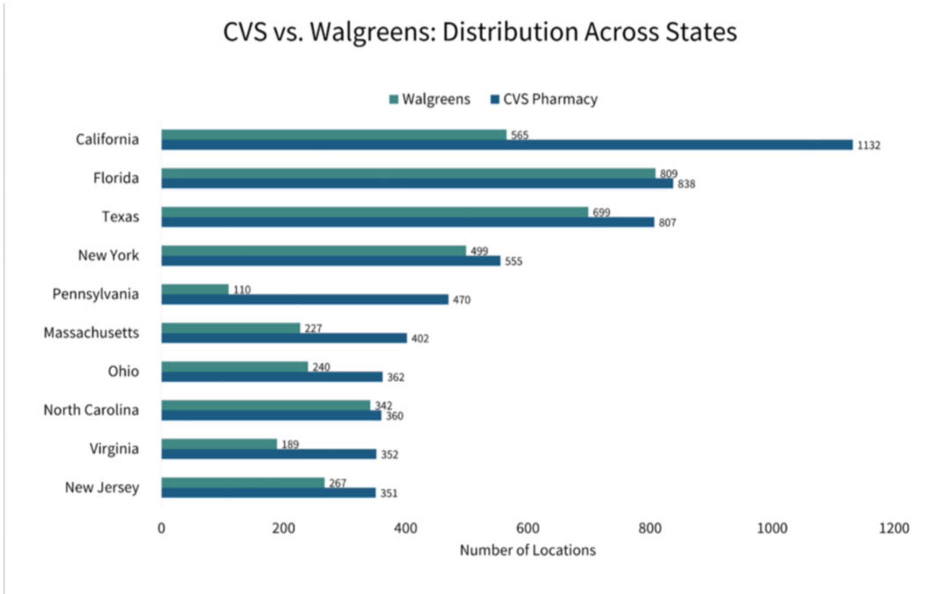

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Database Organization - Entities

- **Mini-world:**
  - Part of a UNIVERSITY environment.
- **Some mini-world entities:**
  - STUDENTs
  - COURSEs
  - SECTIONs (of COURSEs)
  - (academic) DEPARTMENTs
  - INSTRUCTORs
  - PREREQUISITEs
  - GRADEs

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STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

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Relationships Between Entities

- SECTIONs **are of specific** COURSEs
- STUDENTs **take** SECTIONs
- COURSEs **have prerequisite** COURSEs
- INSTRUCTORs **teach** SECTIONs
- COURSEs **are offered by** DEPARTMENTs
- STUDENTs **major in** DEPARTMENTs

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Main Characteristics of the Database Approach

Program-data independence

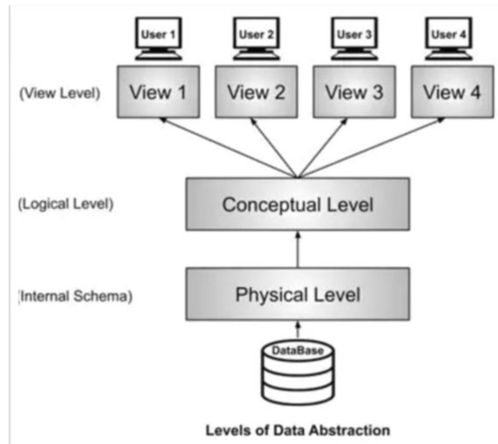
- Allows changing data structures and storage organization without having to change the DB MS access programs.

Data abstraction

- A **data model** is used to hide storage details and present the users with a conceptual view of the database.
- Programs refer to the data model constructs rather than data storage details.

Support of multiple views of the data

- Each user may see a different view of the database, which describes **only** the data of interest to that user.



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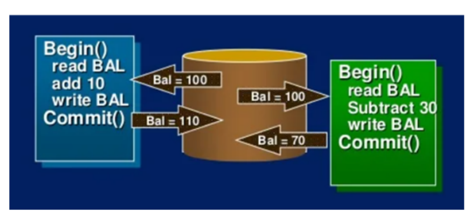
Main Characteristics of the Database Approach

Sharing of data and multi-user transaction processing

- Allowing a set of **concurrent users** to retrieve from and to update the database.
- **Concurrency control** within the DBMS guarantees that each **transaction** is correctly executed or aborted
- **Recovery** subsystem ensures each completed transaction has its effect permanently recorded in the database
- **OLTP** (Online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second.



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Database Schema

The **description** of a database, including:

- database structure
- data types
- data constraints

The UNIVERSITY database

STUDENT

Name	Student_number	Class	Major

COURSE

Course_name	Course_number	Credit_hours	Department

PREREQUISITE

Course_number	Prerequisite_number

SECTION

Section_identifier	Course_number	Semester	Year	Instructor

GRADE

Student_number	Section_identifier	Grade

Database State

The **content** of a database
at some moment in time.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
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119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Data Definition Language (DDL)

- Used by the DBA and database designers to specify the conceptual schema of a database.
- In many DBMSs, the DDL is also used to define internal and external schemas (views).

```
CREATE TABLE EMPLOYEE (
          Fname               VARCHAR(15)         NOT NULL,
          Minit               CHAR,
          Lname               VARCHAR(15)         NOT NULL,
          Ssn                 CHAR(9)             NOT NULL,
          Bdate               DATE,
          Address             VARCHAR(30),
          Sex                 CHAR,
          Salary              DECIMAL(10,2),
          Super_ssn           CHAR(9),
          Dno                 INT                 NOT NULL,

          PRIMARY KEY (Ssn),

          CONSTRAINT EMPSUPERFK
          FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn)
                    ON DELETE SET NULL
                    ON UPDATE CASCADE,

          CONSTRAINT EMPDEPTFK
          FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber)
                    ON DELETE SET DEFAULT
                    ON UPDATE CASCADE
);
```

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Data Manipulation Language (DML)

- Used to specify database retrievals and updates
- Can be applied directly through a **query language**.
- Can be **embedded** in a general-purpose programming language (host language), such as COBOL, C, C++, or Java.

```
INSERT INTO EMPLOYEE VALUES
(Richard','K,'Marini',333445555,'1962-12-30','98
Oak Forest, Katy,TX','M',37000, 987654321,4);
```

```
SELECT Bdate, Address
FROM   EMPLOYEE
WHERE  Fname='John' AND Minit='B' AND Lname='Smith';
```

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Database System Utilities

Tools to:

- Load data files into the database. Includes data conversion tools.
- Back up the database
- Reorganize database file structures
- Monitor performance
- Generate reports
- Other functions, such as sorting, user monitoring, data compression.

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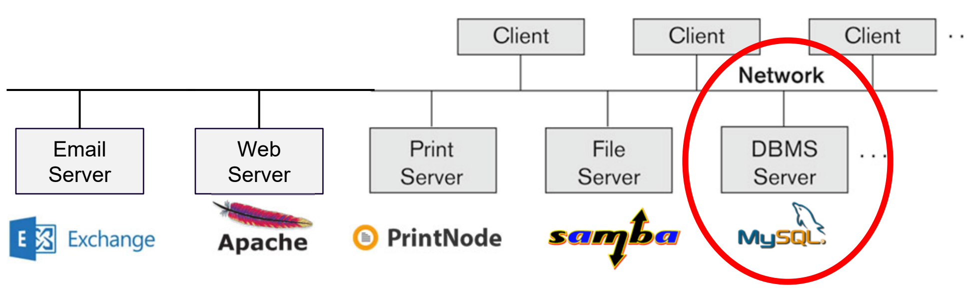
Basic Two Tier Client-Server Architectures

Servers with specialized functions

- Print server
- File server
- DBMS server
- Web server
- Email server

Clients access the specialized servers as needed



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Three-Tier Client-Server Architecture

Common for Web applications